Semantic Subspace Map for Cross-Source Visual Analytics

Yu Dong, Xingyu Liu, Guan Li, Guihua Shan, Dong Tian, Qizhen Yu, Jie Liang, Jian Song, Shiyu Cheng, Zhe Wang, Xuyi Lu, Ruixiao Peng, Yifei Jia, and Ying Zhang

Abstract—Scientific corpora provide vast knowledge across domains and the rise of interdisciplinary research makes multi-domain knowledge discovery increasingly critical. However, extracting and aligning fine-grained semantics from such corpora remains challenging. Existing methods typically emphasize either structural relationships, geometric proximity, or coarse thematic aggregation, but none achieve fine-grained semantic fidelity while supporting interactive, semantics-aware exploration. To bridge this gap, we propose a semantic subspace–driven visual analytics framework that integrates a semantic embedding projection approach with an interactive visual analytics system. The projection approach learns a two-dimensional mapping via multiple losses that preserves semantic embedding similarity and dissimilarity, enabling clear separation of distinct content while maintaining local neighborhoods. The visual analytics system builds on this projection with a map metaphor, supports defining and comparing semantic subspaces, and encodes intra- and cross-subspace knowledge flows; coordinated views and LLM-powered interaction further enable exploration of semantic distributions, subspace-level comparison, and stepwise reasoning. We validate the framework through two case studies and a user study. Results demonstrate its effectiveness and scalability, underscoring the promise of semantic subspace–driven visual analytics for multi-domain knowledge exploration.

Index Terms—Semantic Modeling, Subspace Visual Analytics, Knowledge Exploration

1 Introduction

Interdisciplinary research has become the norm for addressing complex scientific problems. Emerging topics such as energy transition, climate prediction, and bioinformatics routinely cross disciplinary boundaries and demand collaboration between computer science, physics, environmental, and life sciences. Although current digital publishing corpora and retrieval systems have greatly expanded access to domain knowledge, researchers still face fundamental barriers when attempting to interpret and compare findings across domains. The core challenge lies in the heterogeneity of semantics across domains—not only in terminology but also in the functional roles of text segments. Such differences make it difficult for researchers to judge whether another field's problem settings, methodological steps, or conclusions are truly comparable to their own, and the lack of fine-grained semantic representation makes such exploration fragile and costly.

Existing approaches remain inadequate for supporting fine-grained, semantics-aware cross-domain analysis. Citation-based methods [2, 6] (e.g., citespace, knowledge graph) emphasize macroscopic connections among entities, but fail to capture the semantic content expressed at the sentence or paragraph level. For instance, two fields may draw on the same methodological source yet diverge in implementation details or conclusions—subtleties invisible to citation links. Embedding-based projections [26, 33] (e.g., t-SNE, UMAP) map text segments into low-dimensional spaces to visualize proximity, but they mainly optimize geometric neighborhoods rather than preserving core semantic roles (e.g., Background, Method, Conclusion). This often scatters semantically related statements and aggregates unrelated ones, obscuring role-specific flows of knowledge. Topic modeling [15, 25] and automatic summarization provide another route by compressing text into thematic or condensed units, yet these outputs suffer from uneven granularity and overlapping ideas, reducing their reliability for consistent cross-domain alignment. In short, existing approaches either emphasize structure, geometry, or coarse themes, but none ensure fine-grained semantic fidelity while exposing the semantics and relations that are critical for comparative scientific reasoning.

These limitations highlight a critical research gap: *existing methods prioritize either structural relationships, geometric proximity, or coarse thematic aggregation, but none simultaneously achieve fine-grained semantic fidelity and support interactive, context-aware cross-domain exploration.* We argue that a new framework is needed—one capable of preserving semantic consistency in visualization while enabling granular comparative analysis across disciplines. Such a framework should allow researchers to interactively explore semantic subspaces, trace knowledge flows, and align content according to functional roles within scientific discourse, thereby facilitating meaningful cross-domain knowledge exploration.

To address these gaps, we propose a semantic subspace driven projection and visual analytics framework that combines a semantic embedding projection approach with an interactive system. Multi domain papers are first transformed into structured markdown and decomposed into *Minimum Semantic Units* (MSUs), where each MSU expresses a single self contained scientific statement. Each MSU is optionally annotated with a coarse discourse role tag (`pre_subspace`) for stratified analysis. These role tags are used for organizing and reporting results rather than as supervision targets. MSUs are then projected into a shared two dimensional space by learning a lightweight projector on top of fixed sentence embeddings with a contrastive objective and auxiliary constraints, improving semantic fidelity for neighborhood based inspection and cross paper comparison.

On top of the shared projection, we design a hexagonal map metaphor. The plane is discretized by hexagonal tiling, and each hexagonal cell acts as a stable semantic container (HSU) for aggregation and visual encoding. Subspaces are presented as coordinated, role stratified views derived from the same shared projection, enabling focused inspection within a discourse role while preserving a common spatial reference across views. Within and across subspaces, users can externalize semantic connections and trace knowledge flows through interactive path construction and stepwise analysis. The framework is implemented as a visual analytics system that integrates a Semantic Subspace Map and a Stepwise Analysis view, together with large language model support for paper selection and progressive summarization. This design creates a user driven loop of semantic modeling, interactive exploration, and cumulative knowledge narration. The contributions of

- *Yu Dong (dongyu@cnic.cn), Xingyu Liu (xyliu@cnic.cn), Guan Li (liguan@cnic.cn), Guihua Shan (corresponding author, sgh@cnic.cn), Dong Tian (tiandong@cnic.cn), Shiyu Cheng (chengshiyu@cnic.cn), Zhe Wang (zwang@cnic.cn), Xuyi Lu (luxuyi@cnic.cn), Ruixiao Peng (rxpeng@cnic.cn), and Yifei Jia (yfjia@cnic.cn) are with Computer Network Information Center, Chinese Academy of Sciences.*
- *Jie Liang (Jie.Liang@uts.edu.au), Jian Song (jian.song@student.uts.edu.au), and Ying Zhang (ying.zhang@uts.edu.au) are with Faculty of Engineering, University of Technology Sydney.*

Manuscript received xx xxx. 201x; accepted xx xxx. 201x. Date of Publication xx xxx. 201x; date of current version xx xxx. 201x. For information on obtaining reprints of this article, please send e-mail to: reprints@ieee.org. Digital Object Identifier: xx.xxxx/TVCG.201x.xxxxxxx

this paper are threefold:

- We present a learned projection model that improves fine grained semantic fidelity in 2D on top of fixed sentence embeddings, supporting reliable neighborhood inspection and role stratified analysis on a shared coordinate system.
- We introduce an interactive visual analytics system that combines hexagonal map based semantic subspaces with stepwise exploration and large language model assisted summarization, enabling cross view alignment and explicit knowledge flow tracing.
- We demonstrate the framework's effectiveness through two case studies and a user study, showing its generality, usability, and potential for multi domain knowledge discovery.

2 Related Work

2.1 Semantic Embedding and Projection

Semantic embedding is a fundamental task in natural language processing, focusing on the transformation of text—such as words, sentences, or documents—into low-dimensional dense vector representations. These embeddings are designed to capture underlying semantic information, thereby ensuring that texts with similar meanings exhibit closer proximity within the vector space. Early models primarily focused on word-level embeddings. Word2Vec [24] learns word-level embeddings by predicting contextual words, capturing lexical semantic similarities in the resulting vector representations.

Improved embedding models focus on solving semantic representation problems at the sentence or paragraph level, better capturing the overall meaning of longer texts. Transformer architecture has significant advancements in neural text embedding methodologies, largely supplanting earlier statistical and shallow neural approaches [23]. Through bidirectional encoder architecture and masked language modeling pre-training objective, BERT [10] generates context-sensitive, dynamic embeddings, delivering substantial gains across numerous NLP benchmarks. While powerful for token-level tasks, BERT's inherent framework is computationally inefficient for deriving standalone sentence embeddings. Sentence-BERT (SBERT) [28] directly addressed this limitation by introducing a siamese network fine-tuning procedure on natural language inference data. This modification enables the efficient derivation of fixed-dimensional sentence embeddings optimized for semantic similarity comparison via cosine distance, facilitating scalable semantic search.

Subsequent research has focused on refining sentence representations through contrastive learning. SimCSE [12] demonstrated that a simple yet effective unsupervised objective—contrasting embeddings of the same sentence under different dropout masks—could significantly enhance the alignment and uniformity of sentence vectors, outperforming more complex supervised techniques. SimCSE uses self-supervised contrastive learning to construct positive sample sentence embeddings for comparative learning, achieving better semantic embedding results. The self-supervised embedding method can learn high-quality sentence representations without requiring a large amount of annotated data, reducing the dependence of model training on data annotation.

The current state-of-the-art semantic embedding is large-scale, general-purpose embedding models trained on massive datasets of text pairs. Architectures like BGE [35] and E5 [32] leverage advanced contrastive loss functions, instruction-aware training, and Matryoshka Representation Learning to produce highly versatile embeddings. These models exhibit superior performance in critical applications such as semantic search and Retrieval-Augmented Generation.

Embedding projection methods are essential for visualizing high-dimensional data by mapping it into lower-dimensional spaces while preserving structural properties. t-SNE [31] emphasize local similarities by minimizing the Kullback-Leibler divergence between high-dimensional and low-dimensional probability distributions, but they often struggle with global structure preservation and computational scalability for large datasets. UMAP [22] addresses these limitations by leveraging Riemannian geometry and topological data analysis to better balance local and global structure retention, achieving faster performance and improved scalability. Recent advancements also explore manifold-based projections to refine semantic spaces and enhance computational efficiency.

For multi-modal embeddings, ModalChorus [37] introduces a novel approach by integrating metric and non-metric objectives through a Modal Fusion Map, enabling effective visualization and alignment of cross-modal features while mitigating modal gaps. For large-scale network embedding, Shi [29] proposed RandNE, an iterative random projection method that uses Gaussian random projections to preserve high-order node proximities while achieving computational efficiency and distributed computing compatibility. This approach addresses the limitations of traditional methods in handling billion-scale networks by avoiding explicit high-order proximity calculations. In biological data visualization, Liu [21] introduced LOO-map, a framework to assess and improve the reliability of t-SNE and UMAP in single-cell RNA sequencing data. By leveraging leave-one-out perturbation and singularity scoring, LOO-map mitigates distortions in global structure and local neighborhoods without requiring labels.

2.2 Visual Analytics for Scientific Knowledge Exploration

Visual analytics has long supported knowledge discovery in scientific domains by combining computational modeling with interactive exploration. Prior systems have addressed literature analysis, provenance tracking, and sensemaking tasks, as well as domain-specific applications in areas such as climate science and bioinformatics. These systems demonstrate the value of interactive views that allow experts to iteratively refine hypotheses, connect evidence, and document reasoning processes.

Recent work has advanced this agenda by addressing limitations in document representation, interpretability, and integration with intelligent agents. *VADIS* [27] tackles the reliance on static embeddings and opaque similarity maps by introducing a prompt-based attention model for query-specific dynamic embeddings, a relevance-preserving document map, and corpus-level attention visualization. Through iterative interaction, VADIS enables experts to refine queries and achieve more interpretable retrieval results in biomedical literature reviews. Complementing this, *InnovationInsights* [34] explores the dual frontiers of science and technology by linking publications, patents, and researcher profiles. By combining coordinated views such as paper–patent citation flows with a GCN-based prediction model, it supports the identification of promising researchers and the discovery of emerging science–technology linkages.

Parallel efforts integrate large language models (LLMs) into visual analytics workflows. *LEVA* [42] demonstrates how LLMs can serve as intelligent collaborators throughout analysis: assisting onboarding by explaining visual encodings, suggesting alternative perspectives during exploration, and producing narrative summaries of analytic histories. Similarly, *HINTs* [18] combines LLM-based extraction with hypergraph visualization to support sensemaking in large document collections. By aligning prompts to user tasks, modeling documents and keywords as hypergraphs, and offering coordinated cluster, document, and analysis views, HINTs balances automation with interpretability and helps users maintain trust in analytical processes.

Together, these systems illustrate an ongoing shift toward dynamic, explainable, and LLM-augmented approaches for literature analysis and knowledge discovery. Our work extends this line by integrating semantic subspace structures into the visual workflow. In particular, the *Semantic Subspace Map* and *Stepwise Analysis* views support not only navigation and filtering but also the construction of reproducible knowledge traces, enabling users to compare and synthesize insights across papers in a transparent and systematic way.

2.3 Map Metaphor Visualization

Geographic metaphors have been widely adopted in visualization to make abstract structures more interpretable [7, 41]. Knowledge cartography and semantic landscapes, for example, map text or topic distributions into spatial layouts where proximity encodes similarity. Aggregation techniques such as Voronoi diagrams, hexbinning, or semantic clustering have further been used to improve readability and

reveal regional structures.

GMap [11] proposes a map metaphor approach for visualizing general graphs and clusters. It partitions vertices into communities, assigns each cluster to a contiguous region via Voronoi diagrams, and adjusts region sizes for readability, yielding natural map-like layouts with contiguous "countries." Similarly, [36] introduces the Enhanced Hexagon-Tiling Algorithm (EHTA), which represents hierarchical data as contiguous hexagonal regions with irregular geographic-like boundaries, making hierarchical structures more intuitive for non-expert users.

In social media analytics, *D-Map* [4] visualizes ego-centric diffusion patterns as map-like regions on a hexagonal grid, encoding communities, roles, and influence through spatial contiguity and coordinated views. Extending this line of work, *R-Map* [5] proposes to focus relation-centric diffusion by aggregating participants into role-based regions and representing relations as borders and flows, enabling intuitive reasoning about cross-group propagation. In the cyber security domain, [3] introduces *Behavior Map*, which encodes user actions as cities and sessions as trajectories, with size, color, and path thickness capturing action frequency, temporal attributes, and co-occurrence, helping analysts to detect typical patterns and anomalies through a geographic metaphor.

Beyond these applications, Li et al. [20] present *Subspace-Map*, which represents high-dimensional subspaces as "cities" on a hexagonal grid, clusters them into provinces or countries, and uses routes to indicate relationships and transitions, thereby supporting smooth and interpretable exploration of complex subspace structures.

3 Requirement Analysis and Problem Formulation

Our long-term collaborations with experts in engineering simulation, atmospheric modeling, and biology revealed that the main barriers to multi-domain collaboration do not stem from computational scale, but from the difficulty of aligning semantics across domains. Even when researchers examined the same scientific phenomenon, disciplinary conventions produced divergent descriptions and reasoning patterns: some emphasized numerical fidelity, others stressed computational efficiency, while others prioritized interpretability. These heterogeneous perspectives made it difficult to determine whether methods were equivalent, whether assumptions were comparable, or how conclusions were justified.

Through these collaborations, we distilled a common need: to bring heterogeneous research outputs into a comparable semantic frame. Experts called for tools that make fine-grained semantics explicit, allow comparable categories such as background, method, and conclusion to be aligned across disciplines, and reveal how knowledge flows between them. They further emphasized that such a system must not only produce automated mappings but also provide interactive visual support, so that researchers can validate, refine, and explore relations according to domain-specific insights.

Meeting these needs ultimately requires tackling the problem of semantic projection: mapping textual units from diverse disciplines into a two-dimensional space where semantic similarity and difference are faithfully preserved, and where semantic roles and flows can be explicitly visualized. Based on this understanding, we summarize the core challenges as follows:

C1. Granularity and cross-domain consistency. Long sentences or paragraphs often mix multiple ideas, making it difficult to align comparable semantic units across domains. Traditional sentence-level segmentation also fails to capture consistent granularity. Without fine-grained units and consistent categorization, researchers cannot reliably compare different semantics across domains.

C2. Semantic fidelity in projection. General-purpose dimensionality reduction methods preserve geometric neighborhoods but do not explicitly ensure semantic fidelity. As a result, semantically similar text units may be scattered, and semantically different units may be incorrectly grouped. For cross-domain analysis, this undermines the interpretability of projected spaces.

C3. Explicit exploration and knowledge flows. Existing representations of embeddings are dominated by scatterplots, which suffer from overlapping and provide little support for structured exploration. In particular, researchers cannot easily trace how background knowledge leads to methods or how methods support conclusions. The lack of explicit representation of knowledge flows across semantic roles limits knowledge exploration and discovery.

From these challenges, we derived three design goals:

G1. Fine-grained and role-aligned semantic units. The framework should decompose text into Minimum Semantic Units and annotate them with consistent semantic roles (e.g., Background, Method, Conclusion), ensuring that outputs from different disciplines can be compared at the same semantic level.

G2. Projection with semantic fidelity. The projection approach should explicitly preserve semantic similarity and dissimilarity in two-dimensional space, enabling reliable separation of distinct content while aligning comparable units across domains.

G3. Visual encoding of knowledge flows. The visualization should move beyond scatterplots toward aggregated, interpretable representations that explicitly encode intra- and cross-subspace relations, while providing interactive mechanisms for experts to validate, refine, and extend semantic flows.

4 Framework Overview

Building on the challenges (C1–C3) and design goals (G1–G3), we present a semantic subspace–driven framework that integrates a projection approach with an interactive visual analytics system. The framework is designed not only to ensure semantic fidelity in representation, but also to provide an interactive tool where researchers can explore, align, and reason about cross-domain knowledge. Fig. 1 illustrates the overall architecture.

The framework operates as three major stages. First, scientific papers are parsed into Minimum Semantic Units (MSUs), each representing a single scientific semantic. LLM assists this process by segmenting markdown paragraphs, refining sentences, and assigning semantic roles, ensuring fine-grained and role-aligned comparability across domains (G1). Second, a projection model maps these units into two-dimensional space with semantic-aware losses. Here, the LLM also generates paraphrases and semantic triplets to enrich positive/negative pairs, improving the robustness of contrastive learning and enabling reliable separation of distinct content while aligning comparable units (G2). Third, the visual analytics system builds on this projection by aggregating MSUs into Hexagonal Semantic Units (HSUs) and encoding city and country metaphors to strengthen visualizing semantic distribution. Relations across HSUs are encoded as roads and flights, while additional coordinated views and LLM-powered conversational support allow users to filter papers, issue keyword queries, manipulate subspaces, and obtain semantic summaries at the MSU-, HSU-, and link-levels. These mechanisms provide explicit knowledge flows and empower interactive reasoning and validation (G3).

Together, these components form a closed loop: from semantic unit preparation, to semantically faithful projection, to interactive visualization and cumulative knowledge narration. The following sections elaborate on each stage: projection approach (Sec. 5), visual design of semantic subspace map (Sec. 6), and visual analytics system (Sec. 7).

5 Semantic Embedding Projection Approach

5.1 Corpus Preparation and MSU Extraction

We collected 59 representative scientific papers spanning three domains: engine flow-field simulation and optimization, atmospheric numerical modeling and prediction, and computer science (including visualization and AI methods). Each paper is parsed from PDF into OCR-structured markdown using MarkerPDF [8], which preserves paragraph boundaries and positional provenance for downstream tracing.

Cleaning and Meta construction. We remove figure and table captions, as well as caption-like blocks, to avoid fragments whose meaning depends on external visual references. We then clean the OCR-derived text (e.g., normalization of symbols, units, and formatting when applicable) and materialize a structured metadata index, denoted as META (short for *metadata*). META stores paragraph-level records with paper identifier, paragraph index, and character offsets, together

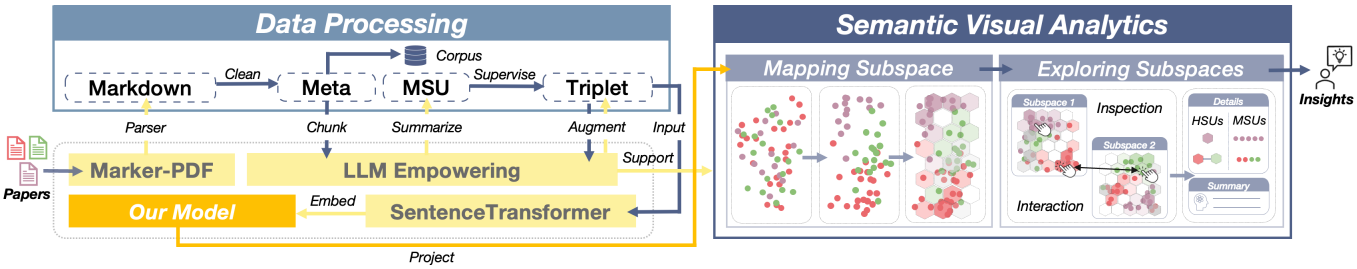

Fig. 1: The framework consists of three major stages from semantic preparation to embedding projection to semantic visual analytics.

with the cleaned paragraph text. It provides an auditable backbone for provenance-aware inspection and document-structure constraints used later in training.

Paragraph-aware chunking with traceable mapping. To ensure stable LLM extraction under context-length constraints while preserving local coherence, we group consecutive paragraphs into context chunks with a fixed maximum length. Chunking is performed on top of META and does not redefine semantic units. For each chunk, we record its identifier, covered paragraph range, and corresponding offsets in META, so every extracted unit can be traced back to its originating paragraphs.

MSU definition and LLM-guided extraction. We define a *Minimum Semantic Unit* (MSU) as a single, self-contained scientific statement that remains interpretable within its local textual context. MSUs are not intended to preserve every original sentence. Instead, they summarize core scientific claims suitable for semantic alignment and projection. Using a fixed decomposition protocol (a structured prompt template with a constrained output schema), we prompt an instruction-following LLM (ChatGPT-4o) to extract MSUs from each chunk. For each MSU, the model outputs (i) the MSU text, (ii) origin pointers to the corresponding paper and paragraph identifiers in META, and (iii) a coarse discourse-role label `pre_subspace` (Sec. 5.2).

Expert supervision and the main MSU corpus. We conduct expert review to remove semantic drift, eliminate duplicates, and correct incomplete or over-fragmented statements. After screening, we obtain a text-only MSU corpus of 2,168 *paper-derived* MSUs from the 59-paper collection. In further augmentation (Sec. 5.3), we additionally extract MSUs from external papers retrieved from arXiv. These auxiliary MSUs are used for training expansion and are not counted in the 2,168 MSUs from the main corpus.

5.2 Discourse-role Labeling for Stratified Analysis

To support stratified visualization and evaluation, we optionally attach each MSU with a coarse discourse-role tag stored as `pre_subspace`. This tag is not used as supervision for training and does not affect the projection model. Instead, it provides a lightweight grouping signal to organize MSUs by their semantic function for analysis (Sec. 5.6). The role schema is configurable and introduced mainly for interpretability of reported results rather than as a fixed taxonomy. For cross-domain scientific papers, we use roles such as `Background`, `Method`, `Experiment`, `Result`, `Limitation`, `Conclusion`, and `Others`. Given a chosen schema, an instruction-following LLM assigns each MSU one role label based on semantic function rather than original section position. In evaluation, we may report only a subset of roles for clarity and stability, while the labeling mechanism itself remains schema-agnostic.

5.3 LLM-augmented Triplet Construction

To supervise projection learning at the semantic level, we construct anchor–positive–negative triplets over MSUs. Triplets are built from two supervision sources: (i) triplets grounded in paper-derived MSUs (from the 59-paper corpus and additional arXiv papers), and (ii) synthetic triplets generated by an LLM as controlled semantic variants. Throughout, we apply expert auditing and drift control to maintain supervision quality.

Candidate triplets from encoder embeddings. We embed paper-derived MSUs using a SentenceTransformer encoder (`bge-large-en-v1.5`), $f : \mathscr{U} \to \mathbb{R}^{1024}$, yielding $\mathbf{h}_i = f(x_i)$. Using cosine similarity in this embedding space, we construct candidate triplets by selecting, for each anchor a, positives from high-similarity neighbors and negatives from low-similarity samples. To increase supervision hardness, we additionally include hard negatives that share topical vocabulary with the anchor but differ in the core claim, and, when identifiable, incompatibility negatives detected by lightweight checks (e.g., mutually exclusive regimes, constraint violations, or inconsistent units).

Expert-audited supervision. We audit and filter candidate triplets under expert supervision to remove noisy matches, semantic drift, and label inversions. This produces a high-precision core set of approximately 9,500 triplets. A small portion is reserved for validation and the remainder is used for training. During mini-batch construction, we optionally draw an additional positive for an anchor from within-paragraph neighbors as a sampling prior. This does not introduce new supervision labels and does not modify the curated triplet set.

Two-source augmentation for semantic coverage. To expand semantic coverage beyond the curated core set, we augment triplets using two complementary mechanisms. We randomly sample 10% of the curated triplets as augmentation seeds and generate ∼30,000 candidate augmented triplets, with roughly 20% from arXiv-driven corpus expansion and 80% from LLM-based synthetic expansion. For the arXiv-driven branch, we query arXiv using anchor-derived keywords, retrieve related papers, and apply the same pipeline as Sec. 5.1 (markdown conversion, cleaning, chunking, and MSU extraction) to obtain auxiliary paper-derived MSUs and corresponding triplets. For the synthetic branch, we use ChatGPT-4o to synthesize additional supervision instances around each sampled anchor. These instances are used only as synthetic triplets and are kept isolated from the held-out evaluation corpus. The generator targets paraphrastic positives, semantics-preserving elaborations, hard negatives, and incompatibility negatives by perturbing claim form, qualifiers, and compatibility constraints while controlling semantic drift.

Quality control and final triplet set. To control semantic drift, we apply de-duplication and rule-based filtering using domain term lists and unit checks to reject candidates that alter key entities, quantities, or physical units. We further conduct stratified human auditing on sampled augmented triplets and refine filtering rules based on observed failure modes. After filtering, we retain approximately 25,000 augmented triplets. Merging them with the curated core set and removing duplicates yields a final supervision set of 34,438 triplets.

5.4 Projection Model and Learning Objectives

Our projection model maps high-dimensional semantic embeddings into a two-dimensional layout space for interpretable visualization. It comprises a SentenceTransformer encoder (`bge-large-en-v1.5`) and a lightweight MLP projector. The encoder is kept fixed during projection learning and only the projector is trained.

Projection network architecture. Let $\mathbf{h}_i = f(x_i) \in \mathbb{R}^{1024}$ denote the embedding of MSU x_i. We learn a projection network $g : \mathbb{R}^{1024} \to \mathbb{R}^2$ implemented as an MLP with three hidden layers of widths 256, 64, and 32, followed by a linear output layer. The projected coordinate is $\mathbf{z}_i = g(\mathbf{h}_i) \in \mathbb{R}^2$. Each hidden layer applies BatchNorm, GELU, and dropout (rate 0.1). The projection is defined in an unconstrained 2D space. Global stability is achieved through the training objective rather than explicit coordinate bounding.

We optimize $g(\cdot)$ with AdamW (learning rate 1×10^{-3}, weight decay 1×10^{-5}). Training uses batch size 128 for 30 epochs with gradient clipping (max-norm 1.0). Only $g(\cdot)$ is updated.

Learning objectives. We train $g(\cdot)$ with a composite objective that preserves semantic ordering, injects weak document structure, discourages local overlap, and stabilizes the global scale of the 2D layout. For any MSU x, let its embedding be $\mathbf{h}(x) = f(x) \in \mathbb{R}^{1024}$ and its 2D coordinate be

$$\mathbf{z}(x) = g(\mathbf{h}(x)) \in \mathbb{R}^2, \tag{1}$$

so that $\mathbf{z}_i = \mathbf{z}(x_i)$ for MSU x_i.

In each mini-batch, we sample N triplets $\{(a_t, p_t, n_t)\}_{t=1}^{N}$, where a_t is an anchor MSU, p_t is its positive, and n_t is its negative. We write $\mathbf{z}_{a_t} = \mathbf{z}(a_t)$, $\mathbf{z}_{p_t} = \mathbf{z}(p_t)$, and $\mathbf{z}_{n_t} = \mathbf{z}(n_t)$. Let $\mathscr{B}$ denote the collection of all triplet elements appearing in the batch, $\mathscr{B} = \{a_t, p_t, n_t\}_{t=1}^{N}$ (thus $|\mathscr{B}| = 3N$, counted with multiplicity).

Within the same mini-batch, we additionally sample two sets of MSU pairs: (i) a structure pair set $\mathscr{P}$ over pairs drawn from $\mathscr{B}$ for hierarchical constraints, and (ii) a repulsion pair set $\mathscr{Q}$ over pairs drawn from $\mathscr{B}$ for overlap prevention. For any pair $(x, x') \in \mathscr{P} \cup \mathscr{Q}$, we use the shorthand $\mathbf{z}_x = \mathbf{z}(x)$ and $\mathbf{z}_{x'} = \mathbf{z}(x')$. All distances are Euclidean in the 2D space, i.e., $\|\mathbf{z}_x - \mathbf{z}_{x'}\|_2$.

The overall objective is

$$\mathscr{L} = \mathscr{L}_{\text{triplet}} + \lambda_{\text{hier}} \mathscr{L}_{\text{hier}} + \lambda_{\text{rep}} \mathscr{L}_{\text{rep}} + \lambda_{\text{st}} \mathscr{L}_{\text{st}}, \tag{2}$$

where $\lambda_{\text{hier}}, \lambda_{\text{rep}}, \lambda_{\text{st}} > 0$ are scalar weights. All hyperparameters are selected on the validation split and then fixed across runs.

Triplet loss (semantic ordering). We enforce that each anchor is closer to its positive than to its negative by a margin $m > 0$:

$$\mathscr{L}_{\text{triplet}} = \frac{1}{N} \sum_{t=1}^{N} \max\left(0,\ \|\mathbf{z}_{a_t} - \mathbf{z}_{p_t}\|_2 - \|\mathbf{z}_{a_t} - \mathbf{z}_{n_t}\|_2 + m\right). \tag{3}$$

Hierarchical pull loss (document-structure coherence). To softly reflect intrinsic document structure, we sample a set of MSU pairs $\mathscr{P}$ from hierarchical relations restricted to the MSUs present in the current batch (e.g., same paragraph, different paragraphs within the same paper, or different papers). We constrain their distances toward hierarchy-dependent target distances:

$$\mathscr{L}_{\text{hier}} = \frac{1}{|\mathscr{P}|} \sum_{(x,x') \in \mathscr{P}} w_{xx'} \left(\|\mathbf{z}_x - \mathbf{z}_{x'}\|_2 - d^*_{xx'}\right)^2. \tag{4}$$

We instantiate a monotonic target scheme over a small set of relation levels and assign targets such that $d^*_{\text{para}} < d^*_{\text{doc}} < d^*_{\text{cross}}$, where $d^*_{\text{para}}, d^*_{\text{doc}}, d^*_{\text{cross}}$ correspond to same-paragraph pairs, cross-paragraph pairs within the same paper, and cross-paper pairs, respectively. Since the global scale is regulated by $\mathscr{L}_{\text{st}}$, the absolute magnitudes of $d^*_{xx'}$ are less important than the relative ordering between relation levels.

Short-range repulsion loss (overlap prevention). Triplet supervision may still allow local overlaps in dense regions. We therefore penalize only pairs that are closer than a short-range radius $\tau > 0$. We sample a label-free set of unordered, non-identical MSU pairs $\mathscr{Q}$ within the current mini-batch and optimize

$$\mathscr{L}_{\text{rep}} = \frac{1}{|\mathscr{Q}|} \sum_{(x,x') \in \mathscr{Q}} \max(0,\ \tau - \|\mathbf{z}_x - \mathbf{z}_{x'}\|_2)^2. \tag{5}$$

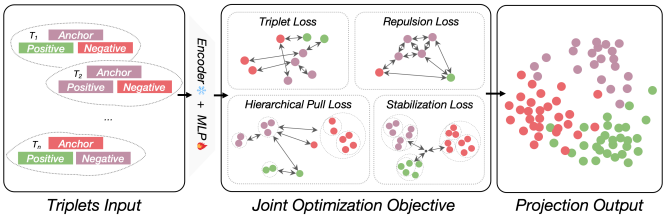

Fig. 2: Overview of the training objective for our semantic projection model. Each MSU triplet (a, p, n) is mapped to a 2D layout through a sentence encoder and a trainable MLP projector. Colors denote the paper of origin (kept consistent across figures), thus a, p, and n may come from different papers. The joint objective combines Triplet Loss for semantic ordering (a closer to p than n), Hierarchical Pull Loss for document structure (e.g., paragraph-level proximity), short-range Repulsion Loss to avoid local overlap, and Stabilization Loss to keep the global scale stable. Dashed enclosures indicate relative distance and grouping cues rather than exact boundaries.

Stabilization loss (scale control). Because the projection is unconstrained, we add a soft stabilizer to prevent unbounded coordinate growth by regularizing the coordinate magnitude of the triplet elements in the batch:

$$\mathscr{L}_{\text{st}} = \frac{1}{|\mathscr{B}|} \sum_{x \in \mathscr{B}} \|\mathbf{z}(x)\|_2^2 = \frac{1}{3N} \sum_{t=1}^{N} \left(\|\mathbf{z}_{a_t}\|_2^2 + \|\mathbf{z}_{p_t}\|_2^2 + \|\mathbf{z}_{n_t}\|_2^2\right). \tag{6}$$

5.5 Training Setup

We split the main corpus by paper to avoid cross-document leakage. We hold out 13 papers (∼900 MSUs) from the 59-paper corpus for evaluation and use the remaining papers for training and validation. All experiments are repeated with five random seeds, and we report mean scores. Baselines (PCA, t-SNE, UMAP) are computed on the same encoder embeddings and averaged over the same five seeds.

5.6 Evaluation Design and Metrics

We evaluate the quality of 2D layouts produced by our learned projector and by PCA, t-SNE, and UMAP as baselines. Our evaluation targets the analysis requirements. The layout should support reliable neighborhood inspection. The layout should remain interpretable and stable for cross-paper auditing.

Evaluation settings. We report results under two settings. (i) *Unified*, where all evaluation MSUs are mapped together. (ii) *Per-role*, where metrics are computed on role-stratified subsets (`pre_subspace`) using the same 2D coordinates. Role labels are used for stratification only and are not training targets.

Evaluation Metrics. We report four task-aligned metrics in the main paper and additional diagnostics in the supplementary material. Silhouette measures paper-level compactness and separation on the 2D coordinates, using paper-of-origin as the reference grouping. NMI measures the agreement between k-means clusters in 2D and paper-of-origin labels, with k set to the number of evaluation papers. To quantify audit-oriented locality, we report Purity@12 and Entropy@12 on 2D neighborhoods with k=12. For each MSU, Purity@12 is the fraction of its 12 nearest neighbors that share the same paper-of-origin label, averaged over all MSUs. Entropy@12 is the Shannon entropy of the paper-label distribution within the 12-neighborhood, averaged over all MSUs. We additionally report Trustworthiness and Continuity in the supplementary material as auxiliary diagnostics.

Reading Table 1. We interpret results along two axes. In the unified setting, higher Silhouette and NMI indicate clearer paper-level separation and stronger agreement with paper-of-origin structure in 2D. Higher Purity@12 and lower Entropy@12 indicate cleaner neighborhoods with less cross-paper mixing, which benefits auditing. In the per-role setting, role-stratified subsets provide a focused view of

Table 1: Comparison using four task aligned metrics (mean over 5 seeds). Silhouette and NMI measure paper separability and label agreement in 2D. Purity@12 and Entropy@12 quantify audit oriented locality of 12 nearest neighbor sets in the 2D layout. Trustworthiness and continuity are reported in the supplementary material for completeness.

Space	Method	Silhouette (↑)	NMI (↑)	Purity@12 (↑)	Entropy@12 (↓)
Unified	PCA	-0.147	0.208	0.230	0.611
	t-SNE	0.074	0.485	0.727	0.200
	UMAP	0.095	0.490	0.684	0.232
	Ours	**0.421**	**0.693**	**0.774**	**0.172**
Experiment	PCA	0.036	0.370	0.502	0.477
	t-SNE	0.062	0.365	0.594	0.387
	UMAP	0.076	0.438	0.555	0.429
	Ours	**0.370**	**0.563**	**0.752**	**0.216**
Limitation	PCA	0.015	0.394	0.470	0.516
	t-SNE	0.070	0.409	0.560	0.443
	UMAP	0.043	0.398	0.515	0.464
	Ours	**0.333**	**0.574**	**0.758**	**0.237**
Conclusion	PCA	0.006	0.363	0.540	0.456
	t-SNE	0.103	0.341	0.650	0.370
	UMAP	0.122	0.430	0.636	0.372
	Ours	**0.454**	**0.658**	**0.802**	**0.186**

neighborhood behavior within each discourse role, while keeping a shared coordinate system. Consistent gains across roles support using coarse roles as an analysis lens without introducing role supervision into the projector. Additional metrics and diagnostics are reported in the supplementary material.

6 Visual Design of Semantic Subspace Map

After obtaining 2D coordinates from the projection model, we design the Semantic Subspace Map to transform raw projections into an interpretable, spatially organized representation of semantic structure. The design adopts a map metaphor, in which clusters of MSUs form coherent regions whose shapes, densities, and boundaries support reasoning about semantic organization and cross-source relationships. This section introduces the layout alternatives, static visual encodings, and interaction-driven encodings that together establish the foundation for the visual analytics system described in Sec. 7.

6.1 Visual Layout Alternatives

We first compare four layout strategies for organizing projected MSUs: scatter, Voronoi, rectangular, and hexagonal based layouts (Fig. 3). In all layouts, MSUs are color-encoded by their source to support cross-paper inspection.

Scatter layouts preserve geometric fidelity but suffer from severe overplotting under multi-source coloring, which obscures fine-grained competition. Voronoi layouts assign each MSU a polygonal region, forming a complete tessellation, but do not provide meaningful aggregation units for expressing multi-source convergence; semantic mixing appears only as subtle color transitions along irregular boundaries, which limits reliable regional comparison. Rectangular and hexagonal layouts mitigate this limitation by aggregating MSUs into regular spatial units. In Fig. 3(c) and (d), units that contain MSUs from multiple sources can be rendered unfilled to explicitly expose areas of semantic overlap. Compared to rectangular grids, hexagonal tiling provides isotropic adjacency, smooth spatial continuity, and adjustable granularity that supports aggregation without distorting neighborhood structure.

For these reasons, we adopt hexagonal tiling as the geometric basis of the visualization. We render the shared projection on a regular hexagonal grid, where each hexagonal cell serves as one Hexagonal Semantic Unit (HSU) for visual encoding. A single parameter r_{hex} controls the tiling resolution (i.e., hexagon size): smaller values yield finer units that preserve sharper spatial variation, while larger values yield coarser units that produce smoother, more consolidated regions.

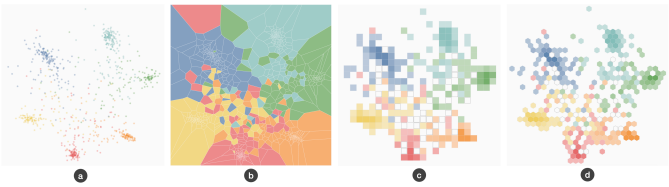

Fig. 3: Comparison of four layout strategies: (a) scatterplot, (b) Voronoi, (c) rectangular, and (d) hexagonal tiling. Hexagonal-based layout provides the most stable and interpretable aggregation for semantic mapping.

This regular tiling provides a stable substrate for the region, boundary, and density encodings described next.

6.2 Static Visual Encodings

On top of the hexagonal aggregation, we define a set of static visual encodings that describe how semantic sources occupy the space, how they intersect, and how dense each area is. These encodings are always present in the map, independent of user interaction.

Exclusive regions. Exclusive regions are defined for each semantic source as connected areas of HSUs whose dominant source equals that semantic source. Each HSU corresponds to one hexagonal cell in the tiling (resolution controlled by r_{hex}), and its dominant source is determined by counting the number of MSUs contributed by each semantic source. Adjacent HSUs with the same dominant source merge into one exclusive region; HSUs that are not adjacent or are separated by overlap are treated as distinct regions. As a consequence, a single semantic source may give rise to several spatially separated exclusive regions, analogous to a mainland and its detached islands on a map. These regions provide a structural view of where a source maintains coherent semantic presence in the projection.

Boundary zones. HSUs that contain MSUs from more than one semantic source are classified as *boundary zones*. Intuitively, if a hexagon aggregates sentences from multiple papers, it is treated as shared rather than belonging to any exclusive region. Boundary zones typically appear along interfaces where semantic sources co-locate or compete for conceptual space. They play a special role in our design: they signal semantic overlap while deliberately avoiding a hard assignment to any single source.

Region color. Each exclusive region can be assigned a distinct color corresponding to its semantic source, allowing users to visually trace where that source occupies the projected space. Boundary zones, in contrast, are initially rendered in a neutral light gray. This neutral treatment makes overlaps easy to identify without implying exclusive ownership, and keeps boundary zones flexible for recoloring in later interactive analysis.

Semantic density. The opacity of each HSU indicates the number of MSUs aggregated into it. Higher opacity marks dense semantic neighborhoods where many MSUs accumulate, suggesting locally coherent or frequently occurring concepts. Lower opacity highlights peripheral or diffuse content. This density cue helps users identify semantic hubs, where MSUs concentrate, as well as gradual transitions in semantic structure across the map.

Boundary encoding. Exclusive regions are enclosed by solid outlines formed by the closed contours of their connected HSUs. Boundary zones are emphasized with dashed outlines. This contrast between solid and dashed boundaries clarifies which areas are uniquely dominated by a single source and which serve as shared or contested semantic territory, making semantic interfaces visually explicit even before interaction.

Together, these static encodings turn the raw projection into a structured thematic map that exposes semantic structure, overlap, and density. The map serves as the visual backbone on which interaction further reveals detailed semantics and user-defined relationships.

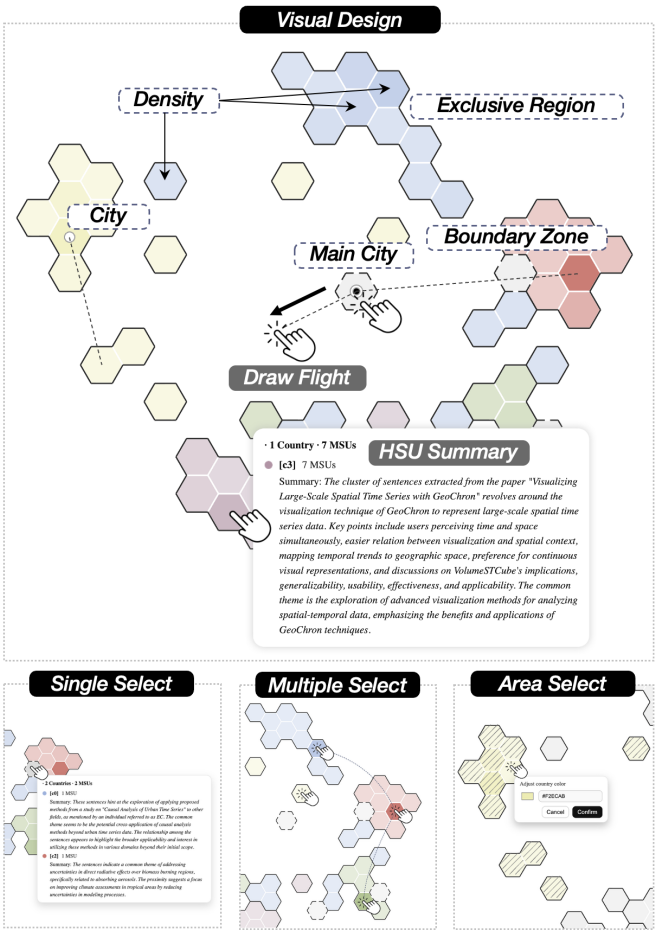

Fig. 4: Top: visual encodings in the Semantic Subspace Map, including density, region, border, city, main city, and HSU summary. Bottom: three selection modes supported in the Semantic Subspace Map View (single select, multiple select, and area select).

6.3 Encodings for Interactions

Beyond the static design, several visual encodings are activated or updated through interaction. These encodings reveal region highlights, semantic summaries, and user-defined semantic paths that extend the map metaphor into an exploratory analytical space (Fig. 4).

Selection and region recoloring. The Semantic Subspace Map supports three complementary selection mechanisms reflected directly through visual encoding. Region selection allows users to drag across the map to select all HSUs within a continuous spatial extent, enabling rapid capture of large semantic neighborhoods or complex boundary patterns. Once selected, exclusive regions and their associated boundary zones can be recolored to assign a unified color to all HSUs originating from a chosen paper or to emphasize a particular overlap configuration. Single selection highlights an individual HSU or exclusive region and reveals its summary panel. Multiple selection allows users to accumulate several HSUs or regions for further comparison within the same analytical step.

HSU summary on hovering. Hovering over an HSU reveals a detail-on-demand summary panel. The panel reports the total number of MSUs in the unit, breaks this count down by semantic source using colored bullets consistent with region colors, and presents an LLM-generated synopsis of the local meaning. This enables semantic interpretation of both single-source and mixed-source HSUs, particularly within dense or overlapping areas, without adding clutter to the static view.

Flights as semantic paths. Users may construct explicit semantic paths, metaphorically referred to as *flights*, to connect HSUs of interest. The interaction employs a two-step gesture: the user first selects a source HSU and then selects one or more target HSUs. Each resulting connection is rendered as a curved path that visually resembles a flight trajectory on a map. Flights may connect HSUs within the same exclusive region or across different regions, enabling user-defined semantic paths that are not restricted to local adjacency or spatial proximity.

Cities and main cities. HSUs that initiate at least one flight are marked as *cities*, indicated by a circular marker at the hexagon center. If an HSU serves as the starting point for multiple outgoing flights, it is designated as a *main city* using a concentric symbol. This encoding highlights semantic hubs that act as departure points for multiple user-defined paths, making branching structures and key reference points immediately recognizable.

These interaction-driven encodings extend the static map into an exploratory workspace. They support fluid transitions between global structure and local meaning, enable users to externalize custom semantic paths by flights, and selectively emphasize regions of interest. Importantly, the interaction semantics are kept consistent across semantic subspaces derived from the shared projection. As a result, flights can connect HSUs across subspaces, region recoloring can be synchronized across panels, and city markers retain their role as stable anchors for interpreting flight endpoints and directions. This cross-subspace consistency enables analysts to reason about semantic relations both within a single subspace and across alternative scopes, supporting richer comparison and subspace-level exploration.

7 Visual Analytics System

We implement a visual analytics system, called Semantic Subspace Map Visual Analytics System (SeSMapVA), which integrates LLM capabilities with spatial semantic visualization to support end-to-end semantic exploration. While the Semantic Subspace Map View serves as the primary analytical workspace, the overall workflow is orchestrated across multiple coordinated views. The LLM is not restricted to a single module; rather, it functions as a unifying semantic engine that supports paper retrieval, MSU extraction, subspace construction, summarization, and cross-subspace comparison. Importantly, the system remains model-agnostic: although realized through ChatGPT-based APIs in the backend, both prompting strategies and LLM providers can be replaced without altering the framework.

7.1 Semantic Subspace Map View

7.1.1 View overview and subspace management

Each semantic subspace is presented as an independent, resizable window that renders its content on the same hexagonal layout and HSU tiling resolution. Each subspace corresponds to a specific semantic filter applied to the shared projection (e.g., only MSUs labeled as *Background* or *Conclusion*). Although the semantic scope differs, all windows initially share identical coordinates and HSU layout, ensuring that corresponding locations remain spatially aligned across subspaces. Windows can be freely dragged and resized on the canvas, allowing users to place subspaces side-by-side for comparison or to compose multiple zoomed subspace views. The window header provides metadata and controls, including the subspace title and action buttons for resetting, duplicating, or closing the window. Titles are editable via double-click so that analysts can document the semantic scope or analytic intent (e.g., "PM2.5 Background" or "Methods of Paper A").

Subspaces are instantiated by combining LLM-driven semantic parsing with user-issued commands. Uploaded or indexed semantic sources are first decomposed into MSUs, and semantic filters (such as roles or paper sets) are inferred and maintained by the back-end. Users then request specific subspaces through conversational commands in the Chat with LLM View (e.g., "show Background and Conclusion subspaces for these papers"). The system applies the requested filter on the shared projection, activates the corresponding MSUs and HSUs, and renders them in new windows.

Window-level operations support iterative refinement. Resetting a window restores its initial state for the current semantic filter, removing temporary selections, recolorings, or user-defined flights while preserving the underlying subspace definition. Duplicating a window creates a new window that inherits the current subspace, enabling users to branch the analysis from an intermediate state and explore alternative hypotheses in parallel. Closing a window removes it from the workspace without affecting other subspaces or the underlying data.

7.1.2 Realization of interaction encodings

Within each window, encoding interactions directly trigger or update the visual encodings defined in Sec. 6. This view operationalizes them through concrete mouse–keyboard gestures.

HSU summary inspection. Hovering over an HSU activates its summary, showing MSU counts, source distribution, and an LLM-generated synopsis.

Selection and recoloring. Single left-click highlights an individual HSU or HSUs connected by flights, while Ctrl–left-click performs multi-selection. Alt-assisted selection activates region-level selection, allowing users to inspect an entire exclusive region together with its associated boundary zones. Once region-level selection is made, a right-click context menu recolors the selected exclusive regions and, optionally, their boundary zones. This updates the region color encoding consistently within the current window and synchronously across other subspace windows that share the same semantic source. The current selection mode is explicitly indicated in the window header through a set of toggle buttons, ensuring that users are always aware of which interaction mode is active before performing any operation.

Flights and city markers. Connection gestures construct *flights* by double-clicks that link a chosen HSU to one or more targeted HSUs. The view renders these trajectories and automatically updates city and main-city markers based on the current set of flights. Flights may also connect HSUs across different subspace windows when views are arranged together.

Saving exploration results. Selections, recolorings, and user-constructed flights can be exported via the *Save* action. The Semantic Subspace Map View thus serves as the workspace where analysts assemble semantic structures of interest, while the Stepwise Analysis View further transforms each saved result into an explicit analytical step with preserved encodings, ready for further decomposition, comparison, and summarization.

7.2 Control Panel

7.2.1 Global system prompt

The system is empowered by LLM to support semantic filtering, subspace construction, and conversational control. The Control Panel exposes a global system prompt that specifies the LLM's responsibilities, including MSU-level semantic assistance, cross-subspace comparison, and structured summarization for visualization. Although the default prompt captures the system's intended behavior, analysts may edit this prompt to inject additional instructions or adapt the model to domain-specific needs.

7.2.2 PDF upload and corpus enrichment

Users may upload scientific papers in PDF format, which are stored and automatically parsed into MSUs through an OCR- and LLM-assisted extraction pipeline. The extracted MSUs are indexed into the knowledge corpus and become immediately available for subspace creation, filtering, and semantic comparison. This direct ingestion mechanism allows experts to expand the knowledge base without external preprocessing and ensures that newly added documents can participate in semantic exploration alongside existing sources.

7.2.3 HSU tiling resolution

The Control Panel exposes r_{hex} to adjust the tiling resolution of the hexagonal map on the shared projection. Updating r_{hex} applies to all open subspace windows synchronously, so cross-window comparisons remain aligned at the same visual granularity.

7.3 Semantic Source Gallery

The Semantic Source Gallery is dynamically populated through natural-language queries issued to the LLM, enabling analysts to retrieve papers by semantic intent rather than by fixed metadata filters. For example, a user may request "show papers on PM2.5 prediction in atmospheric science" or "list all documents discussing deep spatiotemporal models," and this view responds with a curated set of relevant items.

Each retrieved document is displayed as thumbnail cards that combine a visual proxy generated during the PDF–to–MSU parsing pipeline (typically the first extracted figure or a compact composite) with minimal metadata such as title, index number, and domain tag. Hovering over a card reveals additional contextual information, including the markdown-based text summary extracted from the document, allowing analysts to quickly assess whether the source is relevant to the intended semantic scope.

The view supports selection and clearing interactions. Users can click to select one or more sources to specify the sources from which subsequent semantic subspaces will be constructed. Selected papers are marked visually and their MSUs become available to downstream views. The clear action resets the current selection, allowing users to reformulate queries or redefine source sets before initializing new subspaces.

By combining LLM-driven semantic retrieval with visually interpretable document previews, the Semantic Source Gallery provides a flexible and discoverable mechanism for defining the textual basis on which all semantic queries, subspaces, and paths are constructed.

7.4 Chat with LLM View

The Chat with LLM View serves as the visual analytics system's semantic command entrance, allowing users to operate both the Semantic Source Gallery and the Semantic Subspace Map View through natural-language queries. To support efficient interaction, the interface provides contextual quick-access prompts beneath the input field, exposing frequently used operations such as paper retrieval or subspace construction. These prompts guide novice users while still accommodating free-form expert queries.

The supported semantic queries in this view fall into four categories. First, *semantic retrieval* enables users to request papers or MSUs by meaning rather than metadata (e.g., "show papers discussing deep spatiotemporal prediction"), with results populated directly into the Semantic Source Gallery. Second, *semantic summarization* provides concise explanations of selected semantic sources, regions, or HSUs, helping users interpret local semantics without leaving the visual context. Third, *subspace construction commands* allow users to create or refine semantic subspaces via natural-language instructions (e.g., "show Background and Result subspaces for Papers A–C"), forming a fluid bridge between linguistic reasoning and spatial exploration. Finally, *semantic comparison* enables the LLM to articulate similarities, differences, and inconsistencies across multiple sources or subspaces, complementing visual comparison with high-level textual reasoning.

7.5 Stepwise Analysis View

The Stepwise Analysis View transforms each exploration result from the Semantic Subspace Map View into a persistent, structured analytical record. Rather than preserving the entire map, the system decomposes each save into a set of visual cards that capture the semantic evidence selected by users.

7.5.1 Step container

Each save action generates a *step*, shown as a top-level card whose default name mirrors the originating Semantic Subspace Map View window. Users may edit this title to reflect analytical intent or thematic focus. A step groups all semantic elements saved at that moment, providing an organized container for downstream inspection and synthesis.

7.5.2 Decomposition into visual cards

Within each step, the system separates the save into multiple *visual cards*. HSUs selected on the map appear as individual HSU cards, while multi-HSU relations such as flights are represented as dedicated

path cards. Each card includes a flattened visual encoding of the corresponding HSU or flight, including region color, density, and connection styles, which allows users to examine evidence piece by piece rather than revisiting the full map.

7.5.3 MSU-level inspection and filtering

Each visual card carries the MSUs associated with that HSU or path. Users can select one HSU within a visual card to reveal the MSUs aggregated in that HSU. Each MSU is presented with its ID, a one-sentence LLM-generated summary, and its original sentence. A checkbox supports manual filtering: users may exclude MSUs that are spatially aggregated but semantically out of scope, a common requirement given that HSUs emerge from projection and aggregation. This fine-grained interaction allows users to curate the semantic content on which later summaries will be based.

7.5.4 Structured summarization

Based on the selected MSUs, users may invoke the *Summarize* button. The system supplies the LLM with a structured prompt that incorporates the selected MSUs together with their local context and subspace information at the time of saving. When flights span multiple subspaces, this structure-aware prompt guides the LLM to generate a cross-subspace synthesis that respects semantic transitions encoded on the map. In this way, the Stepwise Analysis View not only records interactions but also supports higher-level knowledge construction and reasoning grounded in the user's curated evidence.

8 Case Studies

8.1 Case 1: Visual Analytics for Cross-Domain Knowledge Exploration in Combustion Simulation

We first demonstrate the system through a case study comparing a scramjet combustion simulation paper [30], denoted as P_1, with a visual analytics paper on temporal embedding flows [14], denoted as P_2. This case emphasizes how our design supports semantic distribution, relation construction, and LLM-based synthesis.

Subspace initialization. We begin by defining two semantic subspaces in the *Semantic Subspace Map View*: a *Method Subspace* aggregating sentences about computational approaches and modeling strategies, and a *Conclusion Subspace* aggregating findings about combustion outcomes or semantic interpretation. To reduce overplotting, we adjust the hex radius to $10px$ in the Control Panel, ensuring dense regions of MSUs form contiguous yet interpretable clusters.

Inspecting overlaps and conflicts. Within the *Method Subspace*, we observe overlapping HSUs between P_1 and P_2 around the semantic of *fidelity*. For example, HSU in P_1 emphasizes: *"The computational domain was discretized by a structured mesh with MUSCL interpolation scheme of second-order accuracy to ensure numerical fidelity of turbulence–combustion coupling."* Meanwhile, HSU in P_2 summarizes: *"This method optimizes embedding trajectories to maintain local neighborhood fidelity across temporal slices."* These HSUs are encoded as a conflict region, indicating semantically similar emphases on fidelity despite domain differences.

Link exploration and interaction. Switching to *Route Mode*, we select the semantic path in P_1 from Method (numerical fidelity of simulation) to Conclusion (sustained supersonic combustion): *"The results show that supersonic combustion is sustained with stable ignition characteristics."* Using *Edit Mode*, we create a parallel path in P_2 linking Method (trajectory optimization) to Conclusion (convergent semantic flows): *"Frame embeddings form convergent trajectories that highlight coherent semantic patterns."* Finally, a *flight connection* is constructed between the two subspaces, aligning the shared role of fidelity across combustion modeling and embedding visualization.

Stepwise analysis and synthesis. Clicking *Save* generates a new step in the *Stepwise Analysis View*. The saved step records the connected HSUs, their original sentences, and an LLM-generated synthesis: *"Both papers emphasize fidelity as a core criterion—numerical fidelity in turbulence–combustion coupling (P_1) and semantic fidelity in embedding trajectories (P_2). Despite domain differences, both conclude with stable convergence—sustained supersonic combustion versus convergent semantic flows—suggesting that fidelity-oriented reasoning is transferable across physical and computational domains."* Users can further refine this output by including or excluding MSUs, steering the summarization toward more precise cross-domain articulation.

This case showcases the projection effectively reveals dense and overlapping semantics, the interaction model enables constructing and editing cross-subspace relations, and stepwise analysis provides reusable knowledge artifacts that capture both domain-specific and transferable insights.

8.2 Case 2: Visual Analytics for Multi-Domain Comparative Exploration in Air Pollution Studies

To further demonstrate the generalizability of our system, we conducted a case study on six papers related to air pollution, spanning atmospheric modeling, statistical learning, and visual analytics. Specifically, we include a regression-based PM2.5 prediction paper [17], marked as P_3, a deep learning approach for capturing nonlinear PM2.5 patterns [38], marked as P_4, two studies emphasizing meteorological drivers [1,39], marked as P_5 and P_6, a visual analytics system using frequent subgraph mining for propagation analysis [9], marked as P_7, and a spatio-temporal visualization system for air quality exploration [19], marked as P_8. This case highlights how our system supports large-scale semantic aggregation, comparative subspace construction, and stepwise accumulation of knowledge across heterogeneous approaches.

Subspace initialization. We created three subspaces in the *Semantic Subspace Map View*: a *Method Subspace* aggregating sentences on modeling approaches (e.g., regression, deep learning, frequent subgraph mining, visualization), a *Result Subspace* aggregating performance findings (e.g., prediction accuracy, propagation motifs, correlation structures), and a *Conclusion Subspace* capturing implications such as climate drivers or policy relevance. To ensure legibility, the hex radius was set to $12px$, yielding aggregated HSUs that preserve semantic continuity while avoiding overplotting.

Detecting overlaps and semantic clusters. Using *country mode*, we observed that method-related HSUs formed three semantic regions: (1) data-driven prediction approaches (P_3, P_4), (2) propagation pattern mining (P_7), and (3) spatio-temporal visualization (P_8). Despite methodological diversity, overlaps emerged in the Result Subspace, where HSUs converged on the role of meteorological drivers in explaining variance (P_5, P_6).

Exploring semantic links. Switching to *Route Mode*, we traced paths within individual papers. For example, from Method (deep learning regression) to Result (improved RMSE for Beijing PM2.5 prediction) in P_3, and from Method (pattern mining with MDL compression) to Result (identification of cross-city propagation motifs) in P_7. Using *Edit Mode*, we added intra-subspace roads linking shared emphases on "wind fields" across P_5 and P_6. Finally, *flights* were created to connect heterogeneous approaches across subspaces, e.g., aligning P_4's ability to predict PM2.5 peaks with P_8's identification of episodic propagation events.

Stepwise synthesis. Saving these interactions generated multiple steps in the *Stepwise Analysis View*. Each step captured HSUs, MSU-level summaries, and LLM-generated synthesis. One synthesis stated: *"Prediction models (P_3, P_4) and visual analytics systems (P_7, P_8) converge on the importance of meteorological drivers (P_5, P_6) in shaping PM2.5 dynamics. While models emphasize predictive accuracy, visualization approaches reveal propagation pathways. Together, these findings support hybrid reasoning that combines accuracy-focused learning with interpretive visualization."* Through iterative refinement, users selectively included or excluded MSUs to balance predictive insights with explanatory clarity.

This case demonstrates how our system supports comparative reasoning across six heterogeneous studies. The projection revealed thematic clusters (prediction, propagation, visualization), the interaction model enabled linking methods to results across domains, and the stepwise analysis generated reusable knowledge artifacts. These collectively

validate that our design enhances interpretability, fosters cross-domain synthesis, and facilitates the reuse of knowledge flows in air pollution research.

9 User Study

We conducted a controlled user study to assess whether SeSMapVA supports evidence-grounded semantic exploration and cross-source comparison in a new domain setting. Instead of scoring responses against a single ground-truth answer or benchmarking against another system, we evaluate whether participants can locate evidence, externalize cross-source structure, and verify summaries using the VA system.

9.1 Study Setup

Participants and apparatus. We recruited eight participants (P1–P8) with bioinformatics-related research backgrounds. Participants were graduate researchers or early-career researchers with 4–10 years of research experience (age 31 ± 5 years). All participants were familiar with Circos in prior work, and none had prior experience with IntelliCircos or AuraGenome. None were involved in SeSMapVA development, data preparation, or prompt design. All sessions were conducted in a quiet laboratory environment on the same MacBook Pro (Apple M2 Pro) connected to a 32-inch 5K external monitor.

Materials. We selected three representative papers on circular genome visualization: Circos [16], IntelliCircos [13], and AuraGenome [40]. These papers were not included in the model pretraining or projection training corpus.

All papers were ingested using the same pipeline (PDF → Markdown → MSUs), without introducing domain-specific prompts, templates, or rules. This setting tests whether the same projection-and-subspace workflow transfers beyond the domains used in earlier cases, under participants' realistic prior familiarity with Circos.

9.2 Protocol and Tasks

Protocol. Each session followed a fixed protocol: onboarding and practice, two timed tasks, and a post-task questionnaire. We provided a 30-minute onboarding session with an interaction tutorial and a guided walkthrough. The onboarding covered navigating subspace windows, inspecting HSU summaries, switching selection modes, selecting exclusive regions and boundary zones, constructing flights, and saving steps into the Stepwise Analysis View for MSU-level inspection and summarization. Participants then performed two tasks with upper bounds (T1: 10 minutes; T2: 15 minutes) and could end a task once they considered the collected evidence and saved steps sufficient for their comparison. After the tasks, participants answered a three-item Likert questionnaire (Fig. 5; full item wordings are provided in the supplementary materials).

All activities were completed within SeSMapVA only. External search, personal notes, or additional annotations were not allowed, so that evidence collection and intermediate reasoning could be recorded and inspected consistently within the system. We logged interaction traces (e.g., saves, selections, boundary-zone usage, and flights) and recorded completion time as well as cumulative LLM response latency.[1]

T1: Evidence Localization and Capture ($\leq$ 10 min). Participants located and captured evidence that characterizes each paper's visualization authoring paradigm. They saved steps grounded in HSUs. When overlap or transition semantics emerged during exploration, participants could additionally select boundary zones to capture overlap evidence.

T2: Cross-source Structured Comparison ($\leq$ 15 min). Participants synthesized evidence-grounded comparisons across the three papers using the stepwise workflow. They summarized commonalities and differences across end-to-end authoring pipelines and compared the role of LLMs between IntelliCircos and AuraGenome. Participants could externalize cross-source reasoning by constructing flights linking evidence from different papers, potentially spanning different subspaces.

[1]We define T_{llm} as the summed request-to-response intervals of LLM calls issued during a task. The system remains interactive while requests are pending.

Table 2: User study measures aggregated across participants (median [Q1–Q3]). T_{total} is wall-clock task time. T_{llm} is cumulative LLM response latency (sum of request-to-response intervals). Boundary-zone selections count selected boundary zones. Cross-paper routes count routes whose endpoints come from different papers (N/A for T1).

Measures	Item	T1	T2
Efficiency	T_{total} (min)	*8.1 [7.4–8.8]*	*13.1 [12.2–13.8]*
	T_{llm} (min)	*5.3 [4.7–6.0]*	*8.2 [7.4–9.1]*
	$T_{\text{llm}}/T_{\text{total}}$	*0.65 [0.60–0.70]*	*0.63 [0.58–0.68]*
	N_{llm} (calls)	*16 [14–18]*	*25 [22–28]*
Visuals	Saved steps	*3 [2–4]*	*5 [3–6]*
	Selected HSUs	*5 [4–7]*	*7 [6–9]*
	Selected MSUs	*14 [10–18]*	*18 [14–24]*
	Boundary-zone selections	*2 [1–3]*	*3 [2–4]*
	Cross-paper routes	N/A	*2 [1–3]*

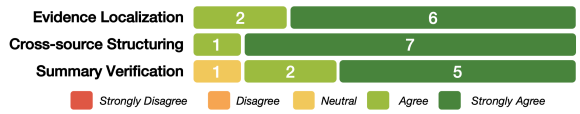

Fig. 5: Post-task subjective ratings (Likert-5) on three dimensions: evidence localization, cross-source structuring, and summary verification.

9.3 Measures

Objective task measures. We report two groups of objective measures from T1 and T2. Efficiency measures include total task time (T_{total}), cumulative LLM response latency (T_{llm}), their ratio ($T_{\text{llm}}/T_{\text{total}}$), and the number of LLM calls (N_{llm}). Together, these measures characterize the contribution of model latency to task duration and the frequency of LLM invocations during analysis. Visuals measure summarize interaction artifacts, including saved steps, selected HSUs, boundary-zone selections, and cross-paper routes.

Subjective measures. We use a three-item Likert-5 questionnaire corresponding to (i) evidence localization, (ii) cross-source structuring, and (iii) summary verification (Fig. 5).

9.4 Results

Task measures. Table 2 reports objective measures aggregated across participants (median [Q1–Q3]). Participants completed both tasks within the predefined upper bounds, with median wall-clock times of 8.1 minutes for T1 and 13.1 minutes for T2. LLM response latency constitutes a substantial portion of task time ($T_{\text{llm}}/T_{\text{total}} \approx 0.63$–$0.65$), reflecting frequent LLM usage within the stepwise workflow. The observed T_{llm} and N_{llm} imply an average single-call latency of approximately 20 seconds, aligning with system runtime observations.

Visual measures. In T1, participants selected a small set of HSUs (median 5) and saved multiple steps (median 3), indicating iterative evidence localization and capture. Boundary-zone selections were also observed (median 2), suggesting that overlap or transition semantics were recorded when relevant. In T2, participants selected more HSUs (median 7) and saved more steps (median 5), consistent with the broader scope of cross-source comparison. Importantly, participants constructed cross-paper routes (median 2), which enabled related HSUs to be treated as a connected evidence structure rather than isolated selections. Within these saved steps, participants further refined evidence at the MSU level (median 18 MSUs), supporting verification and synthesis on top of the connected structure.

Likert ratings. Fig. 5 summarizes the three Likert items to demostrate SeSMapVA's effectiveness and efficiency. All participants reported positive ratings for evidence localization (6 strongly agree, 2 agree) and cross-source structuring (7 strongly agree, 1 agree). For summary verification, most participants reported positive ratings (5 strongly agree, 2 agree) and one participant was neutral.

10 Discussion and Limitation

Discussion. This paper contributes a semantic subspace map–based framework that connects semantic preparation, projection learning, and interactive visual analytics into a single end-to-end pipeline. Across these stages, SeSMapVA keeps semantic units comparable at MSUs, preserves a shared spatial reference through a learned projector, and enables analysts to organize, connect, and verify evidence through subspaces, HSUs, and stepwise analysis. This integration is essential for cross-source comparison: it couples model-side semantic fidelity with interaction-side evidence grounding, so that comparison is performed on explicit evidence rather than on opaque similarity scores or coarse document-level abstractions.

A key methodological outcome is that the learned projection is not merely a geometric embedding, but a task-oriented semantic mapping designed for neighborhood inspection and cross-source auditing. By training a lightweight projector on fixed sentence embeddings with triplet supervision and auxiliary constraints, the projection provides a stable 2D coordinate system in which semantically related MSUs tend to co-locate while dissimilar content remains separable. This property directly supports downstream visual analytics: subspace views inherit the same coordinates, enabling role-stratified inspection without losing cross-view alignment, and HSU aggregation can operate on a consistent spatial substrate rather than on independently re-computed layouts.

In visual analytics, our design highlight the importance of separating unit, connection, and summary. HSUs provide a stable spatial unit for localization and aggregation on the hexagonal map, while boundary zones make cross-source overlap explicit. However, selecting multiple HSUs alone often yields a dispersed set of evidence that naturally leads to separate summaries. Flight metaphor address this gap by connecting related HSUs into routes, enabling analysts to summarize connected evidence as a coherent narrative and to make cross-source structure inspectable. The Stepwise Analysis View then preserves this connected structure together with MSU-level evidence and synthesis, allowing analysts to refine evidence, regenerate summaries, and reuse prior analysis states as reviewable records.

Our evaluations suggest that the framework generalizes as a reusable pipeline rather than as a model-specific system. The same MSU construction, projection, subspace visualization, and stepwise evidence grounding can be applied to new corpora and domains with the same operational logic. As LLM evolve, the framework can adopt alternative providers or local models without changing the core pipeline and interaction design.

Limitation. First, the current framework treats figures only indirectly through captions and local context sentences, and it does not yet support direct figure inspection. A key direction for future work is to incorporate figure-level evidence and align it with MSUs and HSUs, enabling joint reasoning over text and figures under the same evidence and structuring mechanisms.

Second, the framework relatively relies on corpus preparation and LLMs for semantic preparation and for constructing training supervision (e.g., MSU extraction and triplet augmentation). While expert auditing and filtering mitigate semantic drift, scaling to larger corpora will require more systematic quality control and cost-aware implementations, including caching, batching, and adoption of local or compressed models when appropriate. These issues do not undermine the core contributions, but they shape the path toward broader deployment at scale.

11 Conclusion and Future Work

In this paper, we presented a semantic subspace–driven visual analytics framework that bridges semantic embedding and projection, visual design, and interactive exploration for cross-source exploration and insight analysis. By decomposing text into Minimal Semantic Units (MSUs), aggregating them into Hexagonal Semantic Units (HSUs), and projecting them into role-aligned subspaces on a shared coordinate system, the framework preserves fine-grained semantic fidelity and enables direct comparability and summarization across multiple sources.

We formulate projection learning as general approach for transforming semantic embeddings into an ordered 2D semantic space, enabling evidence to be organized, compared, and inspected consistently across sources. We further provide a visual analytics system, called SeSMapVA, supporting end-to-end exploration from evidence localization to cross-source structuring and summary verification. The Semantic Subspace Map, with explicit encodings for regions, boundary zones, flights, and city metaphors, transforms abstract embeddings into interpretable structures and makes cross-source relations inspectable. The Stepwise Analysis View records and reuses semantic routes together with MSU-level evidence and synthesis, turning transient interactions into reviewable analysis records that can be refined and revisited. Case studies and a controlled user study demonstrate that the framework supports rapid, defensible interation and reveals latent semantic contrasts and pathways that are difficult to obtain with standard projection-based exploration. Collectively, these results establish a novel framework for embedding projection and visual analytics that integrates model capabilities with human sensemaking through explicit evidence and structure.

Several directions remain for future work. First, joining multimodal semantics, especially figure, would extend the framework beyond text and strengthen evidence grounding by aligning visual designs with MSUs and HSUs. Second, applying the framework to broader scientific domains and larger source sets will further validate its generality and may motivate domain-specific extensions of subspace organization and route construction. Advancing along these directions will enable the framework to evolve into a robust outcome for multi-domain, multimodal scientific knowledge exploration.

Code Availability

The framework is publicly available on GitHub SeSMap.

References

[1] J. Cai, J. Sulo, Y. Gu, S. Holm, R. Cai, S. Thomas, A. Neuberger, F. Mattsson, M. Paglione, S. Decesari, et al. Elucidating the mechanisms of atmospheric new particle formation in the highly polluted po valley, italy. *Atmospheric Chemistry and Physics*, 24(4):2423–2441, 2024. 9

[2] C. Chen. Searching for intellectual turning points: progressive knowledge domain visualization. *Proceedings of the National Academy of Sciences*, 101(suppl_1):5303–5310, 2004. 1

[3] S. Chen, S. Chen, N. Andrienko, G. Andrienko, P. H. Nguyen, C. Turkay, O. Thonnard, and X. Yuan. User behavior map: Visual exploration for cyber security session data. In *2018 IEEE Symposium on Visualization for Cyber Security (VizSec)*, pp. 1–4, 2018. doi: 10.1109/VIZSEC.2018.8709223 3

[4] S. Chen, S. Chen, Z. Wang, J. Liang, X. Yuan, N. Cao, and Y. Wu. D-map: Visual analysis of ego-centric information diffusion patterns in social media. In *2016 IEEE Conference on Visual Analytics Science and Technology (VAST)*, pp. 41–50, 2016. doi: 10.1109/VAST.2016.7883510 3

[5] S. Chen, S. Li, S. Chen, and X. Yuan. R-map: A map metaphor for visualizing information reposting process in social media. *IEEE Transactions on Visualization and Computer Graphics*, 26(1):1204–1214, 2020. doi: 10.1109/TVCG.2019.2934263 3

[6] Y. Chen, Q. Fan, X. Yuan, Q. Zhang, and Y. Dong. Pgd-gp: a chinese named entity recognition model for constructing food safety standard knowledge graph. *IEEE Transactions on Multimedia*, 2024. 1

[7] F. Cheng, M. S. Keller, H. Qu, N. Gehlenborg, and Q. Wang. Polyphony: an interactive transfer learning framework for single-cell data analysis. *IEEE transactions on visualization and computer graphics*, 29(1):591–601, 2022. 2

[8] DataLab-TO. Marker. `https://github.com/datalab-to/marker`, 2025. GitHub repository. 3

[9] Z. Deng, D. Weng, J. Chen, R. Liu, Z. Wang, J. Bao, Y. Zheng, and Y. Wu. Airvis: Visual analytics of air pollution propagation. *IEEE transactions on visualization and computer graphics*, 26(1):800–810, 2019. 9

[10] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2019. 2

[11] E. R. Gansner, Y. Hu, and S. Kobourov. Gmap: Visualizing graphs and clusters as maps. In *2010 IEEE Pacific Visualization Symposium*

(PacificVis), pp. 201–208, 2010. doi: 10.1109/PACIFICVIS.2010.5429590 3

[12] T. Gao, X. Yao, and D. Chen. Simcse: Simple contrastive learning of sentence embeddings. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 6894–6910, 2021. 2

[13] M. Gu, J. Zhu, Q. Wang, F. Wang, X. Wen, Y. Wang, and M. Zhu. Intellicircos: A data-driven and ai-powered authoring tool for circos plots. In *Computer Graphics Forum*, p. e70118. Wiley Online Library, 2025. 10

[14] Y. Jia, S. Cheng, Y. Dong, G. Li, D. Tian, R. Peng, X. Lu, Y. Wang, W. Yao, and G. Shan. Temporalflowviz: Parameter-aware visual analytics for interpreting scramjet combustion evolution. *arXiv preprint arXiv:2509.04834*, 2025. 9

[15] H. Kim, B. Drake, A. Endert, and H. Park. Architext: Interactive hierarchical topic modeling. *IEEE transactions on visualization and computer graphics*, 27(9):3644–3655, 2020. 1

[16] M. Krzywinski, J. Schein, I. Birol, J. Connors, R. Gascoyne, D. Horsman, S. J. Jones, and M. A. Marra. Circos: an information aesthetic for comparative genomics. *Genome research*, 19(9):1639–1645, 2009. 10

[17] V.-D. Le, T.-C. Bui, and S.-K. Cha. Spatiotemporal deep learning model for citywide air pollution interpolation and prediction. In *2020 IEEE international conference on big data and smart computing (BigComp)*, pp. 55–62. IEEE, 2020. 9

[18] S. Y.-T. Lee and K.-L. Ma. Hints: Sensemaking on large collections of documents with hypergraph visualization and intelligent agents. *IEEE Transactions on Visualization and Computer Graphics*, 31(9):5532–5546, 2025. doi: 10.1109/TVCG.2024.3459961 2

[19] H. Li, H. Fan, and F. Mao. A visualization approach to air pollution data exploration—a case study of air quality index (pm2. 5) in beijing, china. *Atmosphere*, 7(3):35, 2016. 9

[20] J. Li, C. Lai, and X. Yuan. Subspace-map: Interactive visual analysis for subspace data with a map metaphor. *IEEE Transactions on Visualization and Computer Graphics*, pp. 1–15, 2024. doi: 10.1109/TVCG.2024.3368094 3

[21] Z. Liu and Team. Assessing and improving reliability of neighbor embedding methods: a map-continuity perspective. *Nature Communications*, 14(7), 2025. 2

[22] L. McInnes, J. Healy, and J. Melville. Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018. 2

[23] H. Meijer, J. Truong, and R. Karimi. Document embedding for scientific articles: Efficacy of word embeddings vs tfidf. *arXiv preprint arXiv:2107.05151*, 2021. 2

[24] T. Mikolov, K. Chen, G. Corrado, and J. Dean. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*, 2013. 2

[25] D. Palamarchuk, L. Williams, B. Mayer, T. Danielson, R. Faust, L. Deschaine, and C. North. Visualizing temporal topic embeddings with a compass. *IEEE Transactions on Visualization and Computer Graphics*, 2024. 1

[26] R. Peng, Y. Dong, G. Li, D. Tian, and G. Shan. Textlens: large language models-powered visual analytics enhancing text clustering: R. peng et al. *Journal of Visualization*, 28(3):625–643, 2025. 1

[27] R. Qiu, Y. Tu, P.-Y. Yen, and H.-W. Shen. Vadis: A visual analytics pipeline for dynamic document representation and information-seeking. *IEEE Transactions on Visualization and Computer Graphics*, 31(1):1312–1321, 2025. doi: 10.1109/TVCG.2024.3456339 2

[28] N. Reimers and I. Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084*, 2019. 2

[29] B. Shi and T. Weninger. Billion-scale network embedding with iterative random projection. *arXiv preprint*, 2022. 2

[30] Y. Tian, W. Shi, M. Guo, Y. Liu, C. Zhang, and J. Le. Investigation of combustion characteristics in a hydrogen-fueled scramjet combustor. *Acta Astronautica*, 186:486–495, 2021. 9

[31] L. van der Maaten and G. Hinton. Visualizing data using t-sne. *Journal of Machine Learning Research*, 9(86):2579–2605, 2008. 2

[32] L. Wang, N. Yang, X. Huang, B. Jiao, L. Yang, D. Jiang, R. Majumder, and F. Wei. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*, 2022. 2

[33] Y. Wang, L. Chen, J. Jo, and Y. Wang. Joint t-sne for comparable projections of multiple high-dimensional datasets. *IEEE Transactions on Visualization and Computer Graphics*, 28(1):623–632, 2021. 1

[34] Y. Wang, Y. Qian, X. Qi, N. Cao, and D. Wang. Innovationinsights: A visual analytics approach for understanding the dual frontiers of science and technology. *IEEE Transactions on Visualization and Computer Graphics*, 30(1):518–528, 2024. doi: 10.1109/TVCG.2023.3327387 2

[35] S. Xiao, Z. Liu, W. Zhang, B. Su, Y. Huang, Z. Huang, Y. Wen, P. Zhang, Y. Wei, J. Guo, et al. C-pack: Packaged resources to advance general chinese embedding. *arXiv preprint arXiv:2309.07597*, 2023. 2

[36] M. Yang and R. P. Biuk-Aghai. Enhanced hexagon-tiling algorithm for map-like information visualisation. In *Proceedings of the 8th International Symposium on Visual Information Communication and Interaction*, VINCI '15, 6 pages, p. 137–142. Association for Computing Machinery, New York, NY, USA, 2015. doi: 10.1145/2801040.2801056 3

[37] Y. Ye, S. Xiao, X. Zeng, and W. Zeng. Modalchorus: Visual probing and alignment of multi-modal embeddings via modal fusion map. *IEEE Transactions on Visualization and Computer Graphics*, 31(1):294–304, 2025. doi: 10.1109/TVCG.2024.3456387 2

[38] X. Yi, J. Zhang, Z. Wang, T. Li, and Y. Zheng. Deep distributed fusion network for air quality prediction. In *Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining*, pp. 965–973, 2018. 9

[39] X. Yu, M. Zhou, S. Zhu, L. Qiao, J. Li, Y. Ma, Z. Zhang, K. Liao, H. Wang, and J. Z. Yu. Significant secondary formation of nitrogenous organic aerosols in an urban atmosphere revealed by bihourly measurements of bulk organic nitrogen and comprehensive molecular markers. *Atmospheric Chemistry and Physics*, 25(16):9061–9074, 2025. 9

[40] C. Zhang, Y. Dong, Y. Wang, Y. Han, G. Shan, and B. Tang. Auragenome: An llm-powered framework for on-the-fly reusable and scalable circular genome visualizations. *IEEE Computer Graphics and Applications*, (01):1–14, 2025. 10

[41] Y. Zhang, J. Li, and W. Zeng. Latent space map for visual utilization of generated data. *IEEE Transactions on Visualization and Computer Graphics*, 2025. 2

[42] Y. Zhao, Y. Zhang, Y. Zhang, X. Zhao, J. Wang, Z. Shao, C. Turkay, and S. Chen. Leva: Using large language models to enhance visual analytics. *IEEE Transactions on Visualization and Computer Graphics*, 31(3):1830–1847, 2025. doi: 10.1109/TVCG.2024.3368060 2